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Estimation of heavy metals contents and their genotoxic potential in agricultural soil samples of Nangli village, Amritsar (India)

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Abstract : The Current research was aimed to investigate physico-chemical parameters, heavy metal content and genotoxic potential of two soil samples from an agricultural field of Nangli village, Amritsar (India). Samples were collected during cultivation of two different crops i.e. rice (September, 2010) and wheat (March, 2011). The soil samples were characterized for different physico-chemical parameters viz. pH, temperature, electrical conductance, alkalinity, bulk density, water holding capacity, nitrate, phosphate, potassium, sodium, calcium and magnesium. Soil samples were found to be contaminated with different heavy metals (Fe, Mn, Zn, Cd, Ni, Cu and Pb) and also showed high genotoxic potential in *Allium cepa* root chromosomal aberration assay.

Key words :Soil Pollution, Physico-chemical, Heavy metals, Genotoxic, *Allium cepa*.

INTRODUCTION

The contamination of soils by deposition of heavy metals from various sources has attained a great attention worldwide in recent years. The more perilous factor of these contaminants is that they can leach into the ground and surface water systems and make their way into the growing food crops. Contamination of agricultural soils is more prevalent as compared to other soil types as a number of agrochemicals are being used worldwide to obtain more yields of the respective crops. Several reports have clearly documented various human activities as a major cause for heavy metal contamination of the soil ecosystem (Al-Khashman, 2004; Banat *et al.,* 2005; Kasassi *et al.,* 2008) which include mining processes, iron and steel industries, transportation, open disposal of waste, and use of inorganic fertilizers, pesticides on to the agricultural lands (Fuentes *et al.,* 2006, Lado *et al.,* 2008). These pollutants, when present in soil, hinder the nitrogen fixation which is required for the growth of higher plants (Singh *et al.,* 2011; Khan *et al.,* 2013). Therefore, time to time analysis of soil composition is quite essential. Apart from analyzing various contaminants in soils, evaluation of their genotoxic/mutagenic effects in living beings is equally mandatory. Various bioassays have been standardised across the world to reveal the toxicity of different pollutants on living beings and among those, plant bioassays are most recommended because of their durability and easy handling (Rank and Nielsen, 1998). Among various plant bioassays, *Allium cepa* root chromosomal aberration assay has been recommended for genotoxic evaluation of various environmental complex mixtures (Katnoria *et al.,* 2011).

The main objective of present study was to reveal different physico-chemical parameters (pH, temperature, electrical conductance, alkalinity, bulk density, water holding capacity, nitrate, phosphate, potassium, sodium,

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calcium and magnesium) including analysis of essential (Fe, Mn and Zn) as well as toxic heavy metals (Cd, Ni, Cu, Cr and Pb). Their genotoxic potential by applying *Allium cepa* chromosomal aberration assay (*Al*RCAA) in soil samples collected from agricultural field of Nangli village, Amritsar (India). To study the seasonal variations among the various physico-chemical parameters and different contaminants applied, samples were collected during cultivation of two different crops i.e. rice (September, 2010) and wheat (March, 2011).

MATERIALS AND METHODS

Sample collection

For sample collection, soil was dug to the depth of approximately 10-15 cm from the selected site. The samples were allowed to dry for 48 h at room temperature and sieved. Sieved samples were used for measurement of fundamental physico-chemical parameters, heavy metal analysis and genotoxic studies after preparation of extracts. Soil sample collected from Botanical Garden of Guru Nanak Dev University, Amritsar was treated as negative control. The samples were coded as C-r (sample under rice cultivation), C-w (sample under wheat cultivation) and CN (negative control) for all experiments

Physico-chemical analysis

20 g soil was dissolved in 100 ml distilled water (1 : 5 : w/v) and kept on mechanical shaker for 12 h. Solution was filtered through *Whatman* no. 1 filter paper and filtrate was considered as soil extract. pH was measured by pH meter model micro pH system 361 make Systronics, Shimadzu. Alkalinity, bulk density, water holding capacity, calcium and magnesium content were measured using standard protocols (Katnoria *et al.,* 2011). Nitrates and phosphates were estimated using Spectrophotometer model 2202 make Systronics. Potassium and sodium were estimated by flame photometer model CL 26D make ELICO (Trivedy *et al.,* 2002).

Analysis of heavy metals

Analysis of heavy metal in soil and grain samples was carried out in triplicates. For heavy metal analysis, 1 g sample was digested in glass digestion tube of 250 ml with 15 ml nitric acid (HNO_3) at 140°C for 2 h. The content was evaporated to dryness. The dried sample was treated with 3 ml perchloric acid ($HClO_4$) for further oxidation at 240°C for 1h. After digestion, the content was cooled, filtered and made up to 50 ml with double distilled water. The heavy metals were analyzed using Atomic Absorption Spectrophotometer (Shimadzu model AA 6300, Tokyo, Japan). The Hallow cathode lamps (HAMA- MATSU PHOTONICS K.K. JAPAN) of all metals were used as source of radiation.

Estimation of genotoxic potential

For genotoxicity evaluation, two modes of treatments were followed viz. i*n situ* and root dip treatment. Onion bulbs were exposed to soil samples contained in small earthen pots (for *in situ* treatment) and to different concentrations (20%, 40%, 60%, 80% and 100%) of soil extract (1: 2: w/v soil/distilled water) for root dip treatment. After 24-36 h, when roots of 0.5-1 cm length emerged, root tips were washed thoroughly, cut and fixed in Farmer's fluid (3 : 1:: ethanol : acetic acid glacial). Root tips were squashed in aceto-orcein and slides were prepared. The slides were screened under microscope (Olympus CH20i) to study various types of chromosomal aberrations.

RESULTS AND DISCUSSION

A small variation was observed among all physico-chemical parameters of both samples collected under cultivation of rice and wheat (Table 1). Both the samples were found to contain almost neutral pH and high alkalinity. Nitrogen, Phosphate, Potassium, calcium and magnesium contents were not reported to varying as compared to control soil. A number of reports have documented the variation in different physico-chemical parameters in agricultural soils due to application of various contaminants (Katnoria *et al.,* 2008; Doi and Ranamukhaaracchchi, 2009; Kebir and Bouhadjera, 2011).

Among various metals studied, content of heavy metals like Fe and Zn and toxic metals like Cd, Ni and Pb were found to be present in very high concentrations (Table 2). C-r sample was detected with high concentration of toxic metals like Ni (22.69 mg/kg), Cu (24.49 mg/kg) and Pb (130.1 mg/kg). The metal contamination of studied soil area points towards the level of deposition of heavy metals in agricultural soils of Amritsar. Various scientists across the world have reported the presence of very high concentration of heavy metals in agricultural lands. Mico

et al. (2006) recorded high range of Pb (8.9-34.5 mg/kg) in agricultural soils of Spain. Machiwa (2010) studied high concentration of Cd (29.2 µg/g) and Pb (28.5µg/g) in farmland soils of Tanzania. Lado *et al.* (2008) detected concentration of Ni up to 2565 mg/kg in European farm soils. In another report, Kebir and Bouhadjera (2011) documented alarmic concentration of nickel (5.12-34.70 mg/kg) in agricultural soils of Algeria. All these reports point towards the over usage of fertilizers in agricultural land.

Both the samples were found to contain high genotoxic potential in *Allium cepa* test system (Fig 1). C-r and C-w samples induced 12.45% and 12.74% chromosomal aberrations in *in situ* treatment whereas 18.21% and 15.76 % at highest concentration of soil extract (100%) in root dip treatment, respectively. High genotoxicity in C-r sample was probably due to the presence of high concentration of toxic heavy metals like Ni (22.69 mg/kg) and Pb (130.1 mg/kg) as compare to other sample and control. A dose dependent genotoxic response was observed in all the concentrations (20 to 100%) of soil extracts in root dip treatment. This alarmic level of genotoxicity is because of inappropriate usage of xenobiotic chemicals on to the agricultural lands of Amritsar. Similar findings have been reported by our previous studies (Chahal *et al.,* 2012; Katnoria *et al.,* 2011). This study provides clear evidence that soil contaminants raise to various gene defects in living beings.

CONCLUSION

The present study reveals that collected samples were contaminated with one or other heavy metal. However, physico-chemical profile did not show much variation in the samples studied. In addition to this, the samples were also observed to have high genotoxic potential in *Al*RCAA. All these facts clearly show increased level of contamination of agricultural soils of Amritsar and is an alarm to all the farmers as well as the crop consumers in terms of health consequences.

Table 1. Physico-chemical parameters of agricultural soil samples of Nangli Village, Amritsar (India).

Parameter studied	CN	C-r	C-w
pH	6.605±0.05	6.984±1.105	7.6±0.045
Temperature (°C)	34±0.000	34±0.000	34.1±0.000
Electrical Conductance (ms/cm)	0.34±0.000	0.44±0.000	0.27±0.000
Alkalinity (meq/100g)	147.6±0.011	416.7±16.67	466.7±16.67
Bulk density	0.833±0.000	1.487±0.000	0.879±0.000
Water holding capacity (%)	54.74±0.957	51.143±0.326	51.488±0.436
Nitrate (mg/kg)	0.383±0.017	0.12±0.000	0.272±0.004
Phosphate(mg/kg)	0.059±0.000	0.047±0.000	0.045±0.000
Potassium (mg/kg)	0.021±0.007	0.015±0.014	0.020±0.017
Sodium (mg/kg)	0.066±0.000	0.078±0.002	0.122±0.003
Calcium (mg/kg)	0.507±0.014	0.467±0.027	0.494±0.014
Magnesium (mg/kg)	0.657±0.078	0.633±0.333	0.784±0.066

Data shown are Mean±S.E. of three experiments
CN: Negative control; C-r: sample collected under rice cultivation; C-w: sample collected under wheat cultivation

Table 2. Heavy metal content in agricultural soil sample of Nangli Village, Amritsar (India).

Metal studied (mg/kg)	CN	C-r	C-w
Fe	2763±2.792	5844±8.426	5838±3.45
Mn	157.1±16.25	192.9±0.774	201.5±9.508
Zn	69.24±1.002	311.3±6.099	325.0±35.69
Cd	N.D	0.378±0.08	0.262±0.138
Ni	N.D	22.69±5.572	14.73±0.198
Cu	16.27±0.164	24.49±0.269	14.79±0.123
Cr	N.D	N.D	N.D
Pb	N.D	130.1±1.089	13.18±0.742

N.D: Not Detected; Data shown are Mean±S.E. of three experiments
CN: Negative control; C-r: sample collected under wheat cultivation; C-w: sample collected under rice cultivation

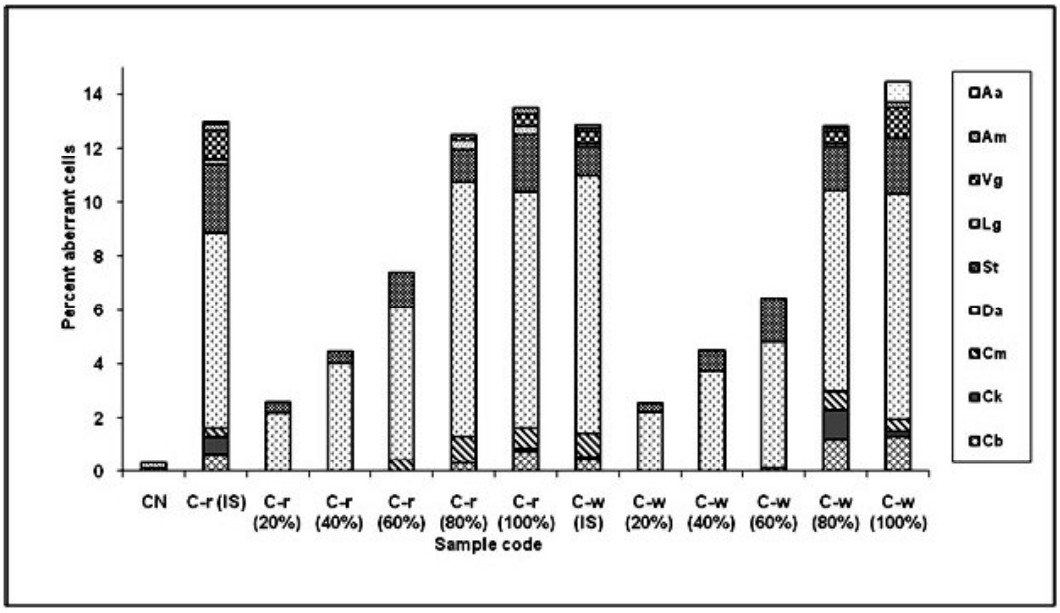


Fig 1. Spectrum of different types of aberrations induced by treatment of root tip cells of *Allium cepa* under *in situ* and root dip treatment.

CN: Negative control; C-r: sample collected under rice cultivation; C-w: sample collected under wheat cultivation; IS: sample under *in situ* treatment.

Aa- Abnormal anaphase; Am- Abnormal metaphase; Vg-Vagrant chromosome/s; Lg Laggard/s; St-Stickiness; Da-Delayed anaphase; Cm-c-mitosis; Ck-Chrmosomal break/s; Cb-Chromatin bridge/s

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